

- BST provided $O(\log n)$ insertion and search for memory segment addresses.
 - In-order traversal returned sorted memory layout: 1000, 2000, 3000, 4000, 5000, 7000, 8000, 9000.
 - Search for 3000: true | Search for 6000: false (not allocated).
- Time Complexity:
- Insert / Search / Delete: $O(\log n)$ average, $O(n)$ worst case.
 - Compared to a simple array, BST performs significantly better for dynamic OS memory operations due to its hierarchical sorted structure.

GitHub Link:

https://github.com/egotpat-gif/DSAIL-Case-Studies/tree/main/DSA_CO1_BST_CASESTUDY